

Normalized cylindric skew Schur polynomials are Lorentzian in two variables, and for one bead in any number of variables

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Abstract

Let $s_{\lambda/\mu}^c(x_1, \dots, x_\ell) = \sum_{\alpha} K_{\alpha}^c x^{\alpha}$ be a cylindric skew Schur polynomial at level n with m beads, and let $N(s_{\lambda/\mu}^c) = \sum_{\alpha} (K_{\alpha}^c / \alpha!) x^{\alpha}$ be its Brändén–Huh normalization. We ask whether $N(s_{\lambda/\mu}^c)$ is Lorentzian. Nonnegativity of coefficients is immediate and M-convexity of the support is known, so the whole question is the Hessian signature condition (L3).

We prove two cases in full. **Theorem 8** gives a sufficient criterion: if the coefficients factor as $K_{\gamma}^c = \prod_t f_t(\gamma_t)$ with each f_t log-concave with interval support, then the normalization is Lorentzian. Since for $m = 1$ one has exactly $K_{\alpha}^c = [\alpha_t \leq n - 1 \ \forall t]$, this settles every one-bead cylindric shape, at every level and in every number of variables (Corollary 9). **Theorem 13** settles $\ell = 2$ for *every* cylindric skew shape, at every level and every m : the constraints on the single intermediate shape decouple into independent intervals, one per bead, so the coefficient sequence is a convolution of interval indicators and is therefore log-concave without internal zeros. The closure of that class under convolution is proved here from Cauchy–Binet.

We also record four *negative* results that close off the routes one would naturally try, and localize the remaining gap precisely: it is exactly the regime $\ell \geq 3$, $m \geq 2$, where bead i at time $t + 1$ is constrained by bead $i + 1$ at time t and the decoupling of Theorem 13 fails. Across 3647 instances (1899 of them winding) no failure of (L3) was found.

1 Setting and the statement

1.1 Cylindric shapes and cylindric skew Schur polynomials

Fix integers $n > m \geq 1$ and work in the bead (Maya) model. A *cylindric shape* at level n with m beads is a bi-infinite strictly increasing sequence $(x_i)_{i \in \mathbb{Z}}$ with $x_{i+m} = x_i + n$; we store it as the tuple $x = (x_1, \dots, x_m)$ with $x_1 < \dots < x_m < x_1 + n$, and we use the periodicity to define x_i for all $i \in \mathbb{Z}$, so in particular $x_0 = x_m - n$ and $x_{m+1} = x_1 + n$. For shapes ν, ρ :

$$\nu \subseteq \rho \iff \nu_i \leq \rho_i \text{ for all } i, \quad (1)$$

$$\rho/\nu \text{ is a cylindric horizontal strip} \iff \nu_i \leq \rho_i < \nu_{i+1} \text{ for all } i, \quad (2)$$

and the size (per period) is $|\rho/\nu| = \sum_{i=1}^m (\rho_i - \nu_i)$.

Given $\mu \subseteq \lambda$ with $d := |\lambda/\mu|$, the *cylindric skew Schur polynomial* in ℓ variables is

$$s_{\lambda/\mu}^c(x_1, \dots, x_\ell) = \sum_{\alpha \in \mathbb{N}^\ell} K_{\alpha}^c x^{\alpha}, \quad K_{\alpha}^c = \# \left\{ \begin{array}{l} \mu = \kappa^0 \subseteq \kappa^1 \subseteq \dots \subseteq \kappa^\ell = \lambda : \\ \kappa^t / \kappa^{t-1} \text{ a cylindric horizontal} \\ \text{strip of size } \alpha_t \ (1 \leq t \leq \ell) \end{array} \right\}. \quad (3)$$

It is homogeneous of degree d . We call K_α^c the *cylindric Kostka numbers*. The polynomial $s_{\lambda/\mu}^c$ is symmetric; equivalently K_α^c depends only on the partition $\text{sort}(\alpha)$. (Verified on all 517 instances of the sweep of §5; we do not use symmetry in either proof below.)

1.2 Lorentzian polynomials

We use the definition of Huh–Matherne–Mészáros–St. Dizier [1, Def. 2.1] (arXiv:1906.09633).

Definition 1. A homogeneous $h = \sum_{|\alpha|=d} c_\alpha x^\alpha \in \mathbb{R}[x_1, \dots, x_\ell]$ is *Lorentzian* if

- (L1) $c_\alpha \geq 0$ for all α ;
- (L2) $\text{supp}(h) = \{\alpha : c_\alpha \neq 0\}$ is *M-convex*: for all $\alpha, \beta \in \text{supp}(h)$ and all i with $\alpha_i > \beta_i$ there is j with $\alpha_j < \beta_j$ and $\alpha - e_i + e_j, \beta - e_j + e_i \in \text{supp}(h)$;
- (L3) for every $\beta \in \mathbb{N}^\ell$ with $|\beta| = d - 2$, the quadratic form $\partial^\beta h$ has at most one positive eigenvalue.

For $d < 2$, (L3) is vacuous.

The *normalization* of $h = \sum_\alpha c_\alpha x^\alpha$ is $N(h) = \sum_\alpha (c_\alpha / \alpha!) x^\alpha$. Our object of study is $N(s_{\lambda/\mu}^c(x_1, \dots, x_\ell))$.

Remark 2 (what is already known, and what is left). (L1) is immediate from (3). For (L2), the support of $s_{\lambda/\mu}^c(x_1, \dots, x_\ell)$ is the set of integer points of a permutahedron slice $P_\lambda \cap \mathbb{Z}^\ell$, and the M-convex exchange axiom for that set is proved (and machine-checked in Lean) in [2]. So the entire remaining content of the question is (L3). Both proofs below are self-contained and re-derive the (L2) they need, so neither depends on [2].

1.3 The normalization turns the Hessian into raw coefficients

The following elementary observation is what makes the problem combinatorial, and it also settles which of two easily-confused inequalities is the right necessary condition.

Lemma 3. Let $h = \sum_{|\gamma|=d} c_\gamma x^\gamma$ and $H = N(h)$. For $\beta \in \mathbb{N}^\ell$ with $|\beta| = d - 2$, the Hessian of the quadratic form $\partial^\beta H$ is

$$M_{ij}^{(\beta)} = \partial^{\beta+e_i+e_j} H = c_{\beta+e_i+e_j}, \quad 1 \leq i, j \leq \ell. \quad (4)$$

That is, the Hessian entries are the raw (un-normalized) coefficients.

Proof. Put $\gamma = \beta + e_i + e_j$, so $|\gamma| = d$. In H the monomial x^γ carries the coefficient $c_\gamma / \gamma!$, and $\partial^\gamma x^\gamma = \gamma!$, while $\partial^\gamma x^{\gamma'} = 0$ for every other γ' of degree d (either some $\gamma'_k < \gamma_k$, killing the term, or $\gamma' = \gamma$). Hence $\partial^\gamma H = (c_\gamma / \gamma!) \cdot \gamma! = c_\gamma$. The Hessian of the quadratic $\partial^\beta H$ at entry (i, j) is $\partial_i \partial_j \partial^\beta H = \partial^{\beta+e_i+e_j} H$. $\square$

Corollary 4. If $N(h)$ is Lorentzian then the raw coefficients satisfy, for all α and all $i \neq j$,

$$c_\alpha^2 \geq c_{\alpha+e_i-e_j} c_{\alpha-e_i+e_j} \quad (\text{RLC})$$

(whenever the shifted indices lie in $\mathbb{N}^\ell$).

Proof. A symmetric matrix with at most one positive eigenvalue has every 2×2 principal submatrix with at most one positive eigenvalue, hence with nonpositive determinant. Apply this to $M^{(\beta)}$ with $\beta = \alpha - e_i - e_j$ and rows/columns i, j : by Lemma 3 the submatrix is $\begin{pmatrix} c_{\alpha+e_i-e_j} & c_{\alpha} \\ c_{\alpha} & c_{\alpha-e_i+e_j} \end{pmatrix}$, whose determinant is $c_{\alpha+e_i-e_j}c_{\alpha-e_i+e_j} - c_{\alpha}^2 \leq 0$. $\square$

Remark 5 (raw versus normalized log-concavity). The same inequality written for the *normalized* coefficients $c_{\alpha}/\alpha!$ is a strictly weaker statement, so (RLC) is the sharp form. Indeed with $p = \alpha + e_i - e_j$, $q = \alpha - e_i + e_j$,

$$\frac{\alpha!^2}{p!q!} = \frac{(\alpha_i!)^2(\alpha_j!)^2}{(\alpha_i+1)!(\alpha_j-1)!(\alpha_i-1)!(\alpha_j+1)!} = \frac{\alpha_i\alpha_j}{(\alpha_i+1)(\alpha_j+1)} < 1,$$

so $c_{\alpha}^2 \geq c_p c_q$ implies $(c_{\alpha}/\alpha!)^2 \geq (c_p/p!)(c_q/q!)$ but not conversely. Both were tested separately in §5; only (RLC) is a consequence of Definition 1.

2 A sufficient criterion: product form

The mechanism in this section is *rank one minus positive semidefinite*, combined with Weyl's inequality. Write $\lambda_1(M) \geq \lambda_2(M) \geq \dots$ for the eigenvalues of a symmetric matrix M , so that “ M has at most one positive eigenvalue” reads $\lambda_2(M) \leq 0$.

Lemma 6. *Let $M = R - P$ with R, P real symmetric, $P \succeq 0$ and $\text{rank } R \leq 1$ with $R \succeq 0$. Then $\lambda_2(M) \leq 0$.*

Proof. Weyl's monotonicity inequality gives $\lambda_2(R - P) \leq \lambda_2(R)$ since $-P \preceq 0$. A rank- ≤ 1 positive semidefinite R has $\lambda_2(R) = 0$. $\square$

Definition 7. A function $f : \mathbb{Z} \rightarrow \mathbb{R}_{\geq 0}$ with finite support is *log-concave with interval support* (for short, PF_2) if $\{k : f(k) > 0\}$ is an interval of integers and $f(k)^2 \geq f(k-1)f(k+1)$ for all $k \in \mathbb{Z}$.

Theorem 8 (product form). *Let $f_1, \dots, f_{\ell}$ be PF_2 , let $d \geq 0$, and put*

$$h = \sum_{|\gamma|=d, \gamma \in \mathbb{N}^{\ell}} \left(\prod_{t=1}^{\ell} f_t(\gamma_t) \right) x^{\gamma}.$$

Then $N(h)$ is Lorentzian.

Proof. Write $c_{\gamma} = \prod_t f_t(\gamma_t)$ and let $[a_t, b_t]$ be the support interval of f_t (if some $f_t \equiv 0$ then $h = 0$ and there is nothing to prove). Then $\text{supp}(h) = \{\gamma \in \mathbb{N}^{\ell} : |\gamma| = d, a_t \leq \gamma_t \leq b_t \forall t\}$.

(L1) is clear.

(L2). Let $\gamma, \delta \in \text{supp}(h)$ and i with $\gamma_i > \delta_i$. Since $|\gamma| = |\delta|$ there is j with $\gamma_j < \delta_j$. Then $\gamma - e_i + e_j$ lies in $\text{supp}(h)$: its i -th entry is $\gamma_i - 1 \geq \delta_i \geq a_i$ and its j -th entry is $\gamma_j + 1 \leq \delta_j \leq b_j$, the other entries are unchanged, and the degree is unchanged. Symmetrically $\delta - e_j + e_i \in \text{supp}(h)$, using $\delta_j - 1 \geq \gamma_j \geq a_j$ and $\delta_i + 1 \leq \gamma_i \leq b_i$.

(L3). Fix $\beta \in \mathbb{N}^{\ell}$ with $|\beta| = d - 2$ and set, for $1 \leq t \leq \ell$,

$$A_t = f_t(\beta_t), \quad B_t = f_t(\beta_t + 1), \quad C_t = f_t(\beta_t + 2).$$

By Lemma 3 and the product form,

$$M_{ij} = \left(\prod_{k \neq i, j} A_k \right) B_i B_j \quad (i \neq j), \quad M_{ii} = \left(\prod_{k \neq i} A_k \right) C_i. \quad (5)$$

Let $Z = \{k : A_k = 0\}$.

Case $|Z| = 0$. Put $P = \prod_k A_k > 0$ and $u_i = B_i/A_i \geq 0$. Then (5) reads $M_{ij} = P u_i u_j$ for $i \neq j$ and $M_{ii} = P C_i/A_i$, so

$$M = P u u^\top - \text{diag}(P(u_i^2 - C_i/A_i))_{i=1}^\ell, \quad u_i^2 - \frac{C_i}{A_i} = \frac{B_i^2 - A_i C_i}{A_i^2} \geq 0,$$

the last inequality being log-concavity of f_i at $\beta_i + 1$. So M is a positive semidefinite rank- ≤ 1 matrix minus a nonnegative diagonal matrix, and Lemma 6 gives $\lambda_2(M) \leq 0$.

Case $|Z| = 1$, say $Z = \{z\}$. For $i \neq j$ with $z \notin \{i, j\}$ the product $\prod_{k \neq i, j} A_k$ contains $A_z = 0$, so $M_{ij} = 0$; and for any i , $\prod_{k \neq i} A_k$ contains $A_z = 0$ unless $i = z$, so $M_{ii} = 0$ for $i \neq z$. Hence all nonzero entries of M lie in row and column z : after permuting z to position 1, $M = \begin{pmatrix} M_{zz} & w^\top \\ w & 0 \end{pmatrix}$. Any vector orthogonal to e_z and to w is in $\ker M$, so $\text{rank } M \leq 2$; and the 2×2 matrix $\begin{pmatrix} M_{zz} & \|w\| \\ \|w\| & 0 \end{pmatrix}$ obtained by restricting to $\text{span}(e_z, w/\|w\|)$ (when $w \neq 0$) has determinant $-\|w\|^2 < 0$, hence one positive and one negative eigenvalue; all other eigenvalues of M vanish. So $\lambda_2(M) \leq 0$. If $w = 0$ then $M = M_{zz} e_z e_z^\top$ and $\lambda_2(M) = 0$.

Case $|Z| = 2$, say $Z = \{z_1, z_2\}$. For $i \neq j$, $\prod_{k \neq i, j} A_k \neq 0$ forces $Z \subseteq \{i, j\}$, i.e. $\{i, j\} = \{z_1, z_2\}$; and $\prod_{k \neq i} A_k \neq 0$ forces $Z \subseteq \{i\}^c$ to contain no zero index, impossible. So the only nonzero entries are $M_{z_1 z_2} = M_{z_2 z_1} =: c$, and M has eigenvalues $\pm|c|$ and 0; thus $\lambda_2(M) \leq 0$.

Case $|Z| \geq 3$. Every product in (5) omits at most two indices and therefore contains some $A_k = 0$, so $M = 0$ and $\lambda_2(M) = 0$. $\square$

Corollary 9 (one bead). *Let $m = 1$, $n \geq 2$, $\mu = (0)$, $\lambda = (d)$ with $0 \leq d$, and $\ell \geq 1$. Then $N(s_{\lambda/\mu}^c(x_1, \dots, x_\ell))$ is Lorentzian.*

Proof. For $m = 1$ condition (2) reads $\nu_1 \leq \rho_1 < \nu_1 + n$, i.e. a cylindric horizontal strip is a step of size $0, 1, \dots, n-1$, and (1) restricts the chain to $[0, d]$. Hence a chain in (3) is precisely a sequence of step sizes $\alpha_1, \dots, \alpha_\ell \in \{0, \dots, n-1\}$ summing to d , and each such sequence occurs once:

$$K_\alpha^c = \prod_{t=1}^\ell \mathbf{1}_{[0, n-1]}(\alpha_t). \quad (6)$$

The indicator $\mathbf{1}_{[0, n-1]}$ is PF₂ (its support is an interval and $1 \geq 1 \cdot 1$, $1 \geq 1 \cdot 0$, $0 \geq 0$ cover all cases), so Theorem 8 applies. $\square$

Remark 10. Equation (6) says that for $m = 1$ the cylindric skew Schur polynomial is the degree- d part of $\prod_{t=1}^\ell (1 + x_t + \dots + x_t^{n-1})$. For $n = 2$ this is e_d , so Corollary 9 contains a two-line proof that the elementary symmetric polynomial e_d is Lorentzian: there $M = J_S - I_S$ for $S = \{i : \beta_i = 0\}$, and $\lambda_2(J_S - I_S) = \lambda_2(J_S) - 1 = -1$. The family (6) *winds* (see §5), so Corollary 9 is not a restatement of the ordinary case.

3 Two variables, every cylindric skew shape

The content of this section is that for $\ell = 2$ the constraints on the single intermediate shape *decouple*, one independent interval per bead.

Proposition 11 (decoupling). *Let $\mu \subseteq \lambda$ be cylindric shapes at level n with m beads, $d = |\lambda/\mu|$. For $1 \leq i \leq m$ set*

$$\ell_i = \max(\mu_i, \lambda_{i-1} + 1), \quad u_i = \min(\lambda_i, \mu_{i+1} - 1), \quad (7)$$

using the periodic convention $\lambda_0 = \lambda_m - n$ and $\mu_{m+1} = \mu_1 + n$. Then the map $\kappa \mapsto (\kappa_1, \dots, \kappa_m)$ is a bijection

$$\{\kappa \text{ cylindric shape} : \kappa/\mu \text{ and } \lambda/\kappa \text{ are cylindric horizontal strips}\} \xrightarrow{\sim} \prod_{i=1}^m [\ell_i, u_i] \cap \mathbb{Z}.$$

Consequently, writing $k_a := K_{(a, d-a)}^c$,

$$\sum_a k_a z^a = \prod_{i=1}^m (z^{\ell_i - \mu_i} + z^{\ell_i - \mu_i + 1} + \dots + z^{u_i - \mu_i}), \quad (8)$$

an empty product of intervals (some $u_i < \ell_i$) meaning $s_{\lambda/\mu}^c(x_1, x_2) = 0$.

Proof. By (1) and (2), a shape κ occurring in (3) with $\ell = 2$ is exactly one satisfying, for every i ,

$$\mu_i \leq \kappa_i, \quad \kappa_i < \mu_{i+1}, \quad \kappa_i \leq \lambda_i, \quad \lambda_i < \kappa_{i+1}, \quad (9)$$

where the first two come from “ κ/μ a horizontal strip” and the last two from “ λ/κ a horizontal strip” (the inequality $\kappa_i \leq \lambda_i$ is the containment $\kappa \subseteq \lambda$). Re-indexing the last constraint as $\kappa_i > \lambda_{i-1}$ and combining, (9) is equivalent to

$$\max(\mu_i, \lambda_{i-1} + 1) \leq \kappa_i \leq \min(\lambda_i, \mu_{i+1} - 1) \quad (1 \leq i \leq m),$$

i.e. to $\kappa_i \in [\ell_i, u_i]$ for each i *separately*: each constraint involves κ_i and the fixed data μ, λ only, never another κ_j . This is the asserted bijection provided every such tuple is in fact a cylindric shape, which we now check. If $\kappa_i \in [\ell_i, u_i]$ for all i then

$$\kappa_i \leq \mu_{i+1} - 1 < \mu_{i+1} \leq \ell_{i+1} \leq \kappa_{i+1},$$

so $\kappa_1 < \kappa_2 < \dots < \kappa_{m+1}$; taking $i = m$ and using $\mu_{m+1} = \mu_1 + n$ gives $\kappa_m \leq \mu_1 + n - 1 \leq \kappa_1 + n - 1 < \kappa_1 + n$. So κ is a cylindric shape, and the periodic extension is consistent.

Finally $a = |\kappa/\mu| = \sum_i (\kappa_i - \mu_i)$ is the sum of the m independent contributions $\kappa_i - \mu_i \in [\ell_i - \mu_i, u_i - \mu_i]$, which is exactly (8). $\square$

We now need that the class PF_2 of Definition 7 is closed under convolution. This is classical (Hoggar); we give a short self-contained proof, because the Cauchy–Binet argument is two lines and we prefer not to rest on an uncited source.

Lemma 12. *For a finitely supported $f : \mathbb{Z} \rightarrow \mathbb{R}_{\geq 0}$ let $T(f)$ be the bi-infinite Toeplitz matrix $T(f)_{ik} = f(k - i)$, $i, k \in \mathbb{Z}$.*

- (i) If all 2×2 minors of $T(f)$ are ≥ 0 , then f is log-concave.
- (ii) If $f = \mathbf{1}_{[p,q]}$ is an interval indicator, all 2×2 minors of $T(f)$ are ≥ 0 .
- (iii) If all 2×2 minors of $T(f)$ and of $T(g)$ are ≥ 0 , the same holds for $T(f * g)$, where $(f * g)(k) = \sum_t f(t)g(k-t)$.

In particular a convolution of interval indicators is log-concave, and it has interval support.

Proof. (i) Take rows $\{i, i+1\}$ and columns $\{k, k+1\}$. The minor is $\det \begin{pmatrix} f(k-i) & f(k+1-i) \\ f(k-i-1) & f(k-i) \end{pmatrix} = f(j)^2 - f(j+1)f(j-1)$ with $j = k-i$, and j ranges over all of $\mathbb{Z}$.

(ii) Take rows $i_1 < i_2$ and columns $k_1 < k_2$, and put $p_* = k_1 - i_1$, $q_* = k_2 - i_2$, $r = k_2 - i_1$, $s = k_1 - i_2$. Then $p_* + q_* = r + s$, and $r > p_*$, $r > q_*$, $s < p_*$, $s < q_*$; so $s < \min(p_*, q_*) \leq \max(p_*, q_*) < r$. The minor is $f(p_*)f(q_*) - f(r)f(s)$. If $f(r)f(s) = 0$ we are done since $f \geq 0$. Otherwise $p \leq s$ and $r \leq q$ (writing $f = \mathbf{1}_{[p,q]}$), whence $p \leq s < p_*$, $q_* < r \leq q$ gives $f(p_*) = f(q_*) = 1$ and the minor is $1 - 1 = 0$.

(iii) $T(f * g) = T(f)T(g)$, since $(T(f)T(g))_{ik} = \sum_j f(j-i)g(k-j) = \sum_t f(t)g(k-i-t) = (f * g)(k-i)$; all sums are finite because f, g have finite support. By Cauchy–Binet, for rows $I = \{i_1, i_2\}$ and columns $K = \{k_1, k_2\}$,

$$\det(T(f)T(g))[I, K] = \sum_{j_1 < j_2} \det T(f)[I, \{j_1, j_2\}] \cdot \det T(g)[\{j_1, j_2\}, K],$$

again a finite sum, and every summand is a product of two nonnegative numbers.

For the last sentence: interval indicators satisfy (ii), hence by (iii) and (i) their convolution is log-concave; and a convolution of nonnegative sequences with interval support has interval support, since $\{k : (f * g)(k) > 0\}$ is the sumset of the two support intervals. $\square$

Theorem 13 (two variables). *For every level $n > m \geq 1$ and every pair of cylindric shapes $\mu \subseteq \lambda$, the polynomial $N(s_{\lambda/\mu}^c(x_1, x_2))$ is Lorentzian.*

Proof. Let $d = |\lambda/\mu|$ and $k_a = K_{(a, d-a)}^c$. By Proposition 11, $\sum_a k_a z^a$ is a product of monomials times interval indicators, so by Lemma 12 the sequence (k_a) is nonnegative, log-concave, and has interval support. If $k \equiv 0$ then $s_{\lambda/\mu}^c(x_1, x_2) = 0$ and there is nothing to prove.

(L1) is clear. (L2): $\text{supp} = \{(a, d-a) : k_a > 0\}$; given $(a, d-a)$ and $(b, d-b)$ in the support with $a > b$, take $j = 2$ (indeed $d-a < d-b$); we need $(a-1, d-a+1)$ and $(b+1, d-b-1)$ in the support, i.e. $k_{a-1} > 0$ and $k_{b+1} > 0$, which holds because $b \leq a-1$ and $b+1 \leq a$ both lie in the support interval $[\min, \max]$ of k . (L3): for $d < 2$ this is vacuous. For $d \geq 2$ and $\beta = (a, b)$ with $a+b = d-2$, Lemma 3 gives

$$M^{(\beta)} = \begin{pmatrix} K_{(a+2, b)}^c & K_{(a+1, b+1)}^c \\ K_{(a+1, b+1)}^c & K_{(a, b+2)}^c \end{pmatrix} = \begin{pmatrix} k_{a+2} & k_{a+1} \\ k_{a+1} & k_a \end{pmatrix},$$

whose determinant is $k_a k_{a+2} - k_{a+1}^2 \leq 0$ by log-concavity. A symmetric 2×2 matrix with nonpositive determinant has eigenvalue product ≤ 0 , hence at most one positive eigenvalue. $\square$

4 Four routes that do not work

Each statement here is checked, not asserted; the point of recording them is that each is a route one would naturally take, and three of the four look like they ought to work.

Proposition 14. *The condition (RLC) of Corollary 4 is not sufficient for (L3), even together with strict positivity of all entries. Explicitly,*

$$M = \begin{pmatrix} 4 & 6 & 4 \\ 6 & 1 & 4 \\ 4 & 4 & 4 \end{pmatrix}$$

is symmetric with strictly positive entries and satisfies $M_{ij}^2 \geq M_{ii}M_{jj}$ for all $i \neq j$ ($36 \geq 4$, $16 \geq 16$, $16 \geq 4$), yet it has two positive eigenvalues.

Proof. The characteristic polynomial of M is $\det(xI - M) = x^3 - 9x^2 - 44x + 16$, whose coefficient sequence $(1, -9, -44, 16)$ has exactly two sign changes; since M is symmetric its eigenvalues are real, so Descartes' rule of signs is exact and M has exactly two positive eigenvalues. (Numerically $\lambda \approx 12.4349, 0.3408, -3.7757$; note $\text{tr } M = 9$ and $\det M = -16$.) $\square$

So no amount of pairwise log-concavity of cylindric Kostka numbers can by itself give (L3); this rules out the “bead hop / exchange” route in its natural form, which produces exactly pairwise exchange inequalities. Two natural strengthenings also fail, and they fail *on the actual Hessians*, not on artificial matrices:

Proposition 15. *Let $n = 4$, $m = 2$, $\mu = (0, 2)$, $\lambda = (2, 4)$, $\ell = 3$, $d = 4$, $\beta = (0, 1, 1)$. Then*

$$M^{(\beta)} = \begin{pmatrix} 2 & 2 & 2 \\ 2 & 0 & 1 \\ 2 & 1 & 0 \end{pmatrix},$$

which has exactly one positive eigenvalue — $\det(xI - M) = x^3 - 2x^2 - 9x - 6$ has coefficient sequence $(1, -2, -9, -6)$ with one sign change, and numerically $\lambda \approx 4.3723, -1, -1.3723$ — and which satisfies (RLC) (the three principal 2×2 minors of M have determinants $-4, -4, -1$), but

- (i) *fails the “Monge” condition $M_{ik}M_{jj} \geq M_{ij}M_{jk}$ ($M_{13}M_{22} = 0 < 2 = M_{12}M_{23}$), and*
- (ii) *fails total nonpositivity of order 2, i.e. nonpositivity of all (not merely principal) 2×2 minors: rows $\{1, 2\}$, columns $\{2, 3\}$ give $M_{12}M_{23} - M_{13}M_{22} = 2 > 0$.*

Consequently neither condition can be the mechanism behind (L3).

Finally, the route through Schur positivity is closed on both sides, and the second reason is the more serious because it survives even where positivity holds.

Proposition 16. *The set of Lorentzian polynomials of a fixed degree is not closed under addition. For instance $(x + y)^2$ and $(2x + y)^2$ are Lorentzian (each has M -convex support and a rank-one positive semidefinite Hessian, so $\lambda_2 = 0$), but their sum $5x^2 + 6xy + 2y^2$ has Hessian $\begin{pmatrix} 10 & 6 \\ 6 & 4 \end{pmatrix}$ with determinant $4 > 0$ and trace $14 > 0$, hence two positive eigenvalues (numerically 13.7082 and 0.2918). Structurally: a Lorentzian quadratic is a rank-one positive semidefinite form minus a positive semidefinite form, and a sum of two rank-one positive semidefinite forms with independent kernels has rank two.*

Remark 17. By Postnikov’s toric Schur theorem, in the range where it applies the Schur expansion of $s_{\lambda/\mu}^c(x_1, \dots, x_\ell)$ has nonnegative (Gromov–Witten) coefficients. One is therefore tempted to deduce Lorentzianity from the ordinary case [1, Thm. 3]. Proposition 16 closes that route: a nonnegative combination of Lorentzian polynomials need not be Lorentzian. Outside that range the route is closed for the cruder reason that Schur coefficients of s^c can be negative — the smallest instance found is $n = 3$, $m = 1$, $\lambda = (3)$, $\ell = 3$, where $s^c = s_{21} - s_{111}$ (13 winding instances with a negative Schur coefficient were found on 2026-09-25). Whatever proves the general case must therefore be polytopal or direct, not a reduction to ordinary Schur polynomials.

5 Computational verification

All code is in `proofs/code-q255/` (`falsify.py`, `sweep.py`, `sweep2.py`, `thmA.py`); the cylindric coefficient generator `weights_counted` is `proofs/code-q254/winding.py` and the checker is `proofs/code-2026-09-19/lorentzian.py`. All arithmetic on (L3) is exact: the number of positive eigenvalues is computed from the integer characteristic polynomial by Faddeev–LeVerrier over $\mathbb{Q}$ plus Descartes’ rule (exact for real-rooted polynomials), never from floating-point eigenvalues.

Instrument controls. The checker passes the untuned positive control of [1, Thm. 3] — $N(s_\lambda(x_1, \dots, x_m))$ Lorentzian — on 16/16 partitions, and *refuses* all three planted negative controls, one each for (L1)/(L2)/(L3). The exact eigenvalue counter was itself wrong on first writing (a mis-indexed Faddeev–LeVerrier recursion); after correction it agrees with its 7 hand controls and with `numpy.linalg.eigvalsh` on 4000/4000 random integer symmetric matrices of size $2 \leq \ell \leq 5$. Lemma 3 was verified symbolically over all β for $\ell = 3, d = 4$.

The main sweep. Over cylindric shapes with $2 \leq n \leq 6$, $1 \leq m \leq n$, $d \leq 9$, $2 \leq \ell \leq 6$:

instances	3647
<i>winding</i> instances (support differs from the unrolled ordinary shape)	1899 (52.1%)
failures of (L2) M-convexity	0
failures of (L3), exact	0
failures of (RLC) on raw coefficients	0

A smaller sweep ($n \leq 5$, $d \leq 7$, $\ell \leq 4$; 517 instances, 228 winding, 44.1%) additionally found 0 failures of the normalized form of (RLC), consistent with Remark 5. The winding fractions are reported because for ℓ even or winding number 0 the model degenerates to the ordinary case, where the answer is known; a sweep confined to those would be a vacuous pass.

Controls on the theorems. The characterization (6) of the $m = 1$ coefficients was checked against the chain recursion for $2 \leq n \leq 7$, $0 \leq d \leq n$, $1 \leq \ell \leq 5$: 165/165 agree. Theorem 8 was checked on 1105 random product-form instances with log-concave f_t : 0 failures. *Negative* control on its hypothesis: among 5573 product-form instances with at least one f_t *not* log-concave, 3604 fail (L3) — so log-concavity of the f_t is load-bearing, not decorative. Proposition 11 was checked by comparing the predicted interval product against both the chain recursion and a brute-force enumeration of intermediate shapes for $2 \leq n \leq 7$, $1 \leq m \leq n$, $d \leq 10$: 3224/3224 shapes agree,

and (k_a) was log-concave with no internal zeros in all 3224. Lemma 12(ii) was checked on $3 \cdot 10^5$ random minors (0 negative) and Lemma 12(iii) on 44409 random PF₂ pairs (0 failures).

Where (L3) is tight. Among the 2238 nonzero Hessians arising from *winding* instances with $m \geq 2$, $\lambda_2(M^{(\beta)}) = 0$ *exactly* in 1637 cases (73.1%) and $\lambda_2 < 0$ in 601 (26.9%). So the conjecture, if true, is true on the boundary of the Lorentzian cone about three times out of four: no proof with slack in it can work. Furthermore the mechanism of Theorem 8 — that the off-diagonal part of $M^{(\beta)}$ is rank one, i.e. $M_{ij}M_{kl} = M_{il}M_{kj}$ for distinct i, j, k, l — holds in 1280 of 1658 Hessians with $\ell \geq 4$ (77.2%), and fails in the remaining 22.8%.

6 Gaps

1. **The main question is open for $\ell \geq 3$ and $m \geq 2$.** This is the whole gap: Corollary 9 covers $m = 1$ for all ℓ , Theorem 13 covers $\ell = 2$ for all m , and $d \leq 1$ is vacuous. Nothing here covers, e.g., $n = 4$, $m = 2$, $\ell = 3$.
2. **The reason the $\ell = 2$ proof does not generalize is identified, and it is structural.** In Proposition 11 the constraints on κ_i involve only the fixed data μ, λ . For a chain of length $\ell \geq 3$ the horizontal-strip condition (2) reads $\kappa_i^{t+1} < \kappa_{i+1}^t$, so bead i at time $t+1$ is constrained by bead $i+1$ at time t : consecutive interior layers couple, and the coefficient sequence is no longer a convolution of independent interval indicators. The coupling is absent for $\ell = 2$ precisely because there is only one interior layer, and absent for $m = 1$ precisely because $i+1 \equiv i$.
3. **The mechanism of Theorem 8 provably does not extend by itself.** Its certificate requires the off-diagonal part of $M^{(\beta)}$ to be rank one, which fails for 22.8% of the winding $m \geq 2$ Hessians tested (§5). This is a statement about *that* certificate only: since $\lambda_2(M) \leq 0$ always admits *some* decomposition $M = \lambda_1 pp^\top - P$ with $P \succeq 0$, what is missing is a combinatorial formula for the rank-one part, not the existence of one. We are careful not to claim more.
4. **Citation constraint, recorded at the point of use.** This paper cites [1] (arXiv:1906.09633) only, for Definition 1 and for the positive control; that entry is **verified-quote**, read at source on 2026-09-19. It deliberately does *not* use any closure theorem of Brändén–Huh (nonnegative linear maps, differential operators, volume polynomials), because the `sources.json` entry for arXiv:1902.03719 carried the grade **deep-read** while its own note read “**POINTER ONLY** ...never checked at source ...Not read”. The grade was corrected to **agent-summary** on 2026-09-26. Several attractive routes to the general case — in particular realizing $N(s^c)$ as a volume polynomial, or transporting Lorentzianity along a nonnegative linear substitution — require those theorems and are therefore *available but not yet citable*. Reading arXiv:1902.03719 at source is the single highest-value next step for this programme.
5. **Not attempted.** The Ehrhart/volume route through the polynomiality of stretched cylindric Kostka coefficients (arXiv:2311.07382, **deep-read**) was identified but not pursued; it appears to need the volume-polynomial theorem of Brändén–Huh, hence gap 4.

References

- [1] J. Huh, J. Matherne, K. Mészáros, A. St. Dizier, *Logarithmic concavity of Schur and related polynomials*, arXiv:1906.09633. Definition 2.1 (Lorentzian) and Theorem 3 (normalized Schur polynomials are Lorentzian).
- [2] Clio, *Cylindric skew Schur polynomials have M-convex support*, `proofs/2026-09-20-c1-cylindric-M-convexity.tex`, Theorem `thm:main`; the exchange axiom is machine-checked (`SortedBridge.sorted_exchange`, sorry-free).